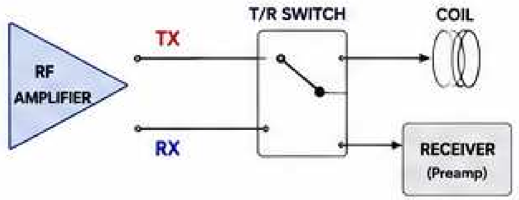
MRI T/R SWITCH & COIL DETUNING – PROTECTING THE RECEIVER

GE MRI

During TRANSMIT the T/R switch connects the RF amplifier to the coil and the coil is DETUNED to protect the low-noise receiver.
During RECEIVE the switch connects the coil to the receiver and the coil is TUNED to detect the ŧŗ tiny MR signal.

1. WHAT IS A T/R SWITCH?

The T/R (Transmit/Receive) switch routes the RF path and isolates the receiver from the high-power transmit pulse.

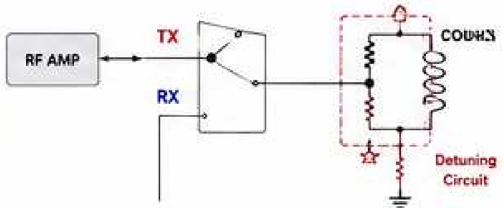


TRANSMIT (TX)
- Connects amplifier to coil
- Isolates receiver

RECEIVE (RX)
- Connects coil to receiver
- Isolates amplifier

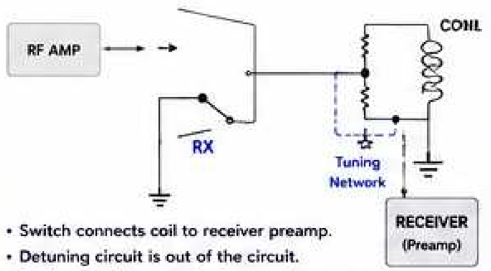
2. HOW IT WORKS (OVERVIEW)

TRANSMIT (Coil Detuned, Receiver Isolated)



- High power from amplifier goes to coil.
- Detuning circuit presents a very low impedance at resonance → coil does not resonate.
- Coil cannot ring or pick up signal.
- Receiver is isolated and protected.

RECEIVE (Coil Tuned, Receiver Connected)



- Switch connects coil to receiver preamp.
- Detuning circuit is out of the circuit.
- Coil is tuned (resonant) at Larmor frequency.
- Tiny MR signal is received with maximum sensitivity.

3. COIL DETUNING – WHY & HOW

WHY DETUNE?
- Prevents large RF voltage induced in the coil during transmit.
- Protects the low-noise preamp from overload and damage.
- Avoids receiver desensitization (ringing/blanking).
- Improves patient safety.
- Reduces amplifier load variations.

HOW DETUNING WORKS

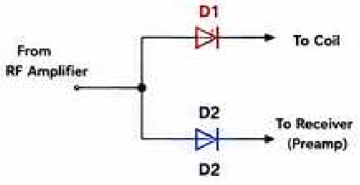
Detuning makes the coil impedance very low at the transmit frequency (f_0).

Common methods:
- PIN diode across capacitor
- PIN diode across resistor
- PIN diode to ground (shunt)
- Trap circuits (notch)
- Active detuning (bias/relays)

Result: Coil Q collapses, it cannot resonate, so it cannot pick up or ring.

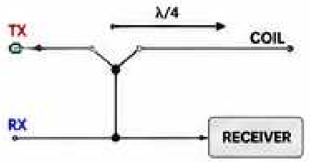
4. TYPICAL T/R SWITCH IMPLEMENTATIONS

PIN DIODE SPDT SWITCH (Most Common)



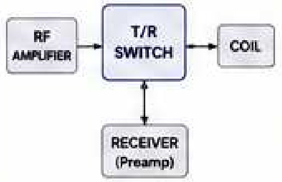
- D1 ON, D2 OFF → TRANSMIT
- D1 OFF, D2 ON → RECEIVE

REFLECTIVE (λ/4) SWITCH



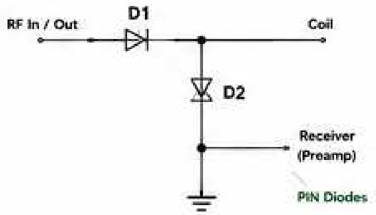
- Quarter-wave line presents high impedance at one port and low at the other.
- No diodes in RF path (low loss).
- Used in many body coils.

MULTI-PORT SWITCH (EXAMPLE)



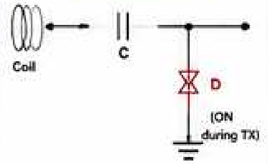
- May include additional ports for:
  - Body coil / local coil routing
  - Multiple receive channels
  - Bypass / load ports

INTERNAL VIEW (PIN DIODE SWITCH)



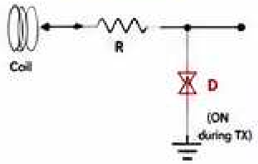
5. COIL DETUNING METHODS (EXAMPLES)

A. PIN DIODE ACROSS CAPACITOR (Parallel Detune)



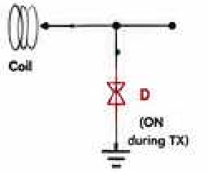
- When diode ON → cap is shorted.
- Coil is heavily damped (low impedance).
- When diode OFF → cap works, coil can resonate.

B. PIN DIODE ACROSS RESISTOR (Resistive Detune)



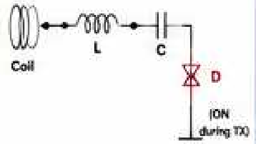
- When diode ON → energy dissipated in R.
- Lowers Q and detunes the coil.
- When diode OFF → coil can resonate.

C. PIN DIODE TO GROUND (Shunt Detune)



- Diode shorts the coil to ground (at RF).
- Very effective detuning.
- When diode OFF → coil tuned.

D. TRAP / NOTCH CIRCUIT (Tuned Trap)



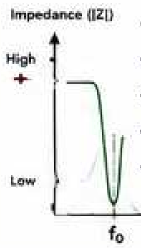
- Trap is tuned to f_0.
- When diode ON → trap absorbs energy.
- When diode OFF → trap out of circuit. Coil can resonate.

6. WHAT HAPPENS WHEN DETUNING FAILS?

Problem	Effect / What You May See
Coil not detuned during transmit	• Very high voltages induced in coil • Preamplifier overload or damage • Zipper / noise / long ringdown • Possible patient heating
Partial detune (weak detune)	• Reduced receiver sensitivity • Blanking / dead time increase • Artifacts in first few lines of k-space
T/R switch stuck in TX	• No receive signal (black images) • Receiver not connected
T/R switch stuck in RX	• No transmit power or very low power • System transmit fault
Poor isolation	• Leakage of high power to preamp • Elevated noise floor, reduced SNR

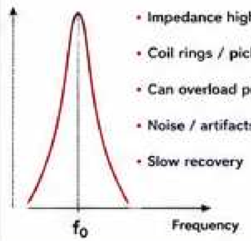
7. GOOD vs POOR DETUNING (IMPACT)

GOOD DETUNING (During TX)



- Impedance very low at f_0
- No resonance, no ringing
- Preamplifier protected
- Clean transmit pulse
- Fast recovery

POOR / NO DETUNING (During TX)



- Impedance high at f_0
- Coil rings / picks up
- Can overload preamp
- Noise / artifacts
- Slow recovery

Note: A properly detuned coil should present < 5 – 10 Ω at f_0 during transmit.

9. TYPICAL SPECIFICATIONS (GUIDELINE)

Parameter	Typical Value / Guideline
Isolation (TX to RX)	> 80 dB (100 W systems) > 90 dB (High power systems)
Insertion Loss (TX path)	< 0.5 dB
Insertion Loss (RX path)	< 0.5 dB
Detune Impedance at f_0 (TX state)	< 5 – 10 Ω
Switching Speed	< 10 µs (typ.)
Power Handling (TX path)	Up to 2 – 10 kW peak (system dependent)
Diode Type	PIN diodes (low capacitance, high power)

9. CHECKS & TEST POINTS (SERVICE)

- Verify DC bias to diodes (TX/RX states).
- Check diode continuity (forward/reverse).
- Measure isolation with network analyzer.
- Measure detune impedance at f_0.
- Verify switching speed (scope if available).
- Check for arcing / heating / discoloration.
- Inspect coax connectors and solder joints.
- Validate receive noise with test load.
- Run system QA after repairs.

Always follow ESD precautions and safety procedures. High RF power is present during test.

10. QUICK REFERENCE

	TX STATE (Transmit)	RX STATE (Receive)
T/R Switch	TX port connected	RX port connected
Coil State	Detuned (non-resonant)	Tuned (resonant)
Detune Diode	ON (conducting)	OFF (open)
Path	AMP → Coil	Coil → Preamp
Receiver	Isolated / Protected	Connected / Active
Purpose	Deliver power safely	Detect tiny MR signal

IMPORTANT NOTES

- Never bypass or disable detuning circuits.
- Use only approved components and diodes.
- Incorrect repairs can damage the receiver or RF amplifier.
- Verify performance with measurements, not assumptions.

COMMON DIAGNOSIS CLUES

- Zipper noise at start of echo → Detune issue
- Long ringdown → Detune issue
- Low SNR across all coils → Detune / preamp issue
- No transmit power → Switch / bias / control issue
- No receive signal → Switch / cable / preamp issue
- Localized artifact → Coil element or trap failure

ABBREVIATIONS

TX = Transmit
RX = Receive
T/R = Transmit / Receive
PIN = Positive Intrinsic Negative (diode)
Q = Quality Factor
f_0 = Larmor Frequency

THE BOTTOM LINE

A properly operating T/R switch and coil detuning protect the receiver, ensure patient safety, and enable high-quality MR images.

Proper Switching + Effective Detuning = Receiver Protection, High SNR, Fast Recovery, and Reliable System Performance.